

Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohammed Razia Alangir Banu
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Roll Number	2520030138
Branch	CSE
Course Outcome	4

Case Study Topic: UrbanAI – Intelligent Smart City Infrastructure Optimization Platform

Work Done Summary:

In this case study, Dijkstra's Algorithm was implemented to find the shortest and fastest routes for emergency service vehicles navigating a smart city grid. A weighted graph was utilized where critical city intersections are represented as nodes and dynamic road travel times or distances serve as edges. The algorithm dynamically calculated the minimum travel costs from the central UrbanAI command center to all surrounding response zones. The optimal pathways were retrieved and displayed efficiently using a priority queue structure. This system minimizes critical emergency response latency and highlights the vital role of graph algorithms in modern smart city transportation management.

Implementation Details:

- Created a weighted graph structure representing city intersections and dynamic road network distances.
- Represented the central AI command station and various target municipal zones as graph vertices.
- Stored real-time travel impedance and route lengths as weighted edges.
- Applied Dijkstra's Algorithm to systematically determine the single-source shortest paths.
- Utilized a min-Priority Queue to optimize distance and cost update calculations.
- Displayed the minimum path cost from the command station to each municipal zone.
- Generated complete shortest-route telemetry records for smart city logistics management.

Result: Screenshots / Output

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===== UrbanAI - Emergency Response Routing System =====
```

```
Shortest Distance from Command Center:
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AI Command Center = 0 km  
Municipal Zone A = 4 km  
Municipal Zone B = 5 km  
Municipal Zone C = 9 km  
Municipal Zone D = 8 km
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Route Optimization Completed Successfully.
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Time Complexity:

Time Complexity: $O((V + E) \log V)$

Conclusion:

The Emergency Response & Traffic Flow Routing System successfully identified the most efficient and shortest paths across the urban network using Dijkstra's Algorithm. By modeling city infrastructure as a weighted graph paired with a priority queue execution, the platform guarantees rapid, live routing updates. This directly reduces response delays, demonstrating how advanced graph data structures drive intelligent decision-making in high-stakes smart city frameworks.

GitHub Link:

<https://github.com/marahrudaydeepak-ctrl/DSA-2>

Student Signature	Faculty Signature